Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/30/2017
PI Name:	O'Banion, Kerry M.D., Ph.D.		
Project Title:	Impact of Space Radiation Induced Neuroinflammation on Alzheimer and Parkinson Disease Pathology		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kerry_obanion@urmc.rochester.edu	**Fax:**	FY 585-756-5334
PI Organization Type:	UNIVERSITY	**Phone:**	585-275-5185
Organization Name:	University of Rochester		
PI Address 1:	Box 603		
PI Address 2:	601 Elmwood Ave		
PI Web Page:			
City:	Rochester	**State:**	NY
Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	11/13/2012	**End Date:**	10/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2016 per S. Monk/LaRC (Ed., 12/11/15)		
Key Personnel Changes/Previous PI:	September 2014: Lisa Opanashuk recently took a position with the FDA and is no longer a University of Rochester Faculty member. We are identifying a new collaborator to assist with analysis of the PD mouse brain sections. September 2013: Sean Hurley was listed in this record as a Co-Investigator; however, Sean left the University prior to the start of this grant and was probably inadvertently included because of the delay between the initial application and the actual funding date. I removed his name from the Co-Investigator section.		
COI Name (Institution):	Olschowka, John (University of Rochester Medical Center) Williams, Jacqueline (University of Rochester Medical Center)		
Grant/Contract No.:	NNX13AC33G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The brain reacts to injury by activation of glial cells and expression of inflammatory mediators in a process called neuroinflammation. Neuroinflammation occurs in the setting of age-associated neurodegenerative diseases such as Alzheimer's (AD) and Parkinson's (PD) and numerous studies suggest a contributory role to disease pathogenesis. We have previously shown that space radiation, at doses approaching those that will be experienced by astronauts on long missions to the moon and Mars, elicits brain neuroinflammation. Therefore we propose to investigate the possibility that space radiation influences pathological changes associated with neurodegenerative diseases. Our studies will utilize three different mouse models of the two most prevalent age-associated neurodegenerative diseases (Alzheimer's and Parkinson's) to assess the effects of space radiation on disease-associated pathology and behavioral changes as well as indices of neuroinflammation and oxidative stress. Findings from these studies will help to refine risk estimates for future space explorers and may offer possible strategies for mitigation of such risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Information was gained about specific mouse models used for neurodegenerative disease, including the timing of onset and sex differences in the models. This information will be useful for others utilizing these same models for investigation of Alzheimer and Parkinson diseases.
Task Progress:	Aim 1. 3x Tg AD Mice. We arranged for production of aged triple transgenic AD mice (hereafter referred to as 3xTg) by Jackson Laboratory and in October and November 2013 successfully carried out three separate irradiation campaigns during NASA Space Radiation Laboratory (NSRL) NSRL-13C. The specific dates and beams were: 1) 300 MeV/µ Silicon at 10 and 100 cGy on October 31; 2) 600 MeV/µ Iron at 10 and 100 cGy on November 15; and 3) 200 cGy mixed protons mimicking a solar flare on November 14. For each ion and energy 10 male and 10 female mice were irradiated. Two set of sham irradiated (control) mice with 10 males and 10 females in each set were included in the experiment, for a total of 140 transgenic mice. Mice were approximately 7 months of age at the time of irradiation and were shipped back to Rochester to be aged. Notably, the mice were sacrificed at 16-17 months of age, which is younger than planned due to unexpected deaths, particularly in males, that were not correlated to irradiation. In our earlier studies using a different AD mouse model (APPswe/PS1dE9) we found behavioral deficits and increased amyloid beta plaque accumulation following 100 cGy 56Fe particles (1 GeV/µ; Cherry et al., 2012. PLoS one 7:e53275). Although our earlier study showed radiation dependent deficits in two behavioral tests, we were not able to replicate these findings with the 3xTg mice. To determine whether 3xTg mice show exacerbation of amyloid pathology with irradiation, we carried out immunohistochemical staining of tissue sections with an antibody that recognizes amyloid beta (Aß). Although Aß plaques were readily observed in female 3xTg mice, we did not detect Aß deposits in male mice in any brain region. Across three brain regions there was only one significant change observed: an increase in cortical Aß load in female mice exposed to 10 cGy iron (600 MeV/µ). ELISA measures for human Aß1-40 and Aß1-42 revealed substantial variability within each group; indeed, ANOVA for all samples analyzed showed no significant effect of radiation. Multiple studies in transgenic mice suggest that increased Aß production or deposition is associated with increased tau phosphorylation and tangle pathology, perhaps by Aß induced neuroinflammation. Therefore we also examined the effects of radiation on tau phosphorylation, an antibody against an epitope that is prominently phosphorylated in AD brain. Similar to the findings with Aß, phospho-tau staining was increased only in the female mice exposed to 10 cGy iron. To examine inflammatory changes we also carried out immunohistochemical staining of ICAM-1, an endothelial activation marker that is upregulated in irradiated mouse brain. As with Aß and tau, there was increased staining for ICAM-1 in brains of female mice exposed to 10 cGy iron; interestingly, male mice also showed increased staining with iron exposure. Taken together, these findings in the 3xTg mouse model of AD illustrate differences between sexes with regard to presence of pathology. It also seems clear that the various ions used (proton, silicon, and iron) gave distinct responses. Aim 2. Mutant Parkin Mice. We contracted with Jackson laboratories to perform a rapid derivation of Parkin-Q311X mice with a goal of 80 mice (40 male, 40 female) harboring the disease model transgene and 80 control littermates. They were shipped to Brookhaven National Laboratory (BNL) in mid April and utilized during the Spring 2014 NSRL run (NSRL14A). The specific dates and beams were: 1) 300 MeV/µ Silicon at 50 cGy on May 1; 2) 600 MeV/µ Iron at 50 cGy on May 12; and 3) 100 cGy mixed protons mimicking a solar flare on May 13. In each instance we irradiated groups of 10 male and 10 female mice representing two genotypes (Parkin-Q311X or wild-type) with each ion and had one set of sham-irradiated control mice. These animals were approximately 8 months of age at the time of irradiation and were shipped to Rochester. Prior to sacrifice at 18 months of age, mice underwent behavioral tests that are sensitive to basal ganglia dysfunction. Some patterns emerged from these data. For example, WT (wild type) female mice irradiated with Silicon (50 cGy) showed clearly increased behavioral activity across many measures. Interestingly, no such pattern was seen with male mice and no other radiation type had such an effect. When activity is analyzed as total activity, male Parkin mice showed reduced activity vs. WT mice on a number of measures. Interestingly, radiation seemed to affect the overall performance of WT mice, bringing them closer to the activity levels seen in Parkin mice. To investigate neurodegeneration related changes in this model of PD, we carried out an immunohistochemical stain for tyrosine hydroxylase to label dopaminergic neurons and their projections. Blinded densitometric analysis of striatal TH staining showed evidence for reduced fiber staining in male, but not female mutant transgenic mice that was not influenced by radiation. The finding that Parkin mice showed reduced striatal dopamine fiber staining compared with WT mice is consistent with the reduced activity seen in this sex. Aim 3. mRNA Analyses. RNA samples were prepared from the combined hippocampus and forebrain of 3xTg AD mice and the striatum of Parkin/WT mice and subjected to cDNA synthesis followed by multi-array quantitative real-time PCR for 48 separate genes. The mRNAs examined included selected chemokines, cytokines, cell-activation markers, enzymes involved in inflammatory mediator production, and responses to oxidative stress. Initial analysis of gene expression profiles does not support a clear picture of increased inflammation and oxidative stress following radiation. Results from these studies are being incorporated into two manuscripts, one describing the 3xTg mouse experiments and the second reporting results from the Parkin mouse studies. With regard to future studies investigating the effects of space radiation on neurodegenerative disease, our challenges with the two models studied here suggest that other models be used in the future. In particular, models with a more robust/reproducible phenotype, such as the APP/PS-1 AD mouse, are likely to produce clearer results.

Bibliography Type:	Description: (Last Updated: 01/07/2026)
Articles in Peer-reviewed Journals	Sweet TB, Hurley SD, Wu MD, Olschowka JA, Williams JP, O'Banion MK. "Neurogenic effects of low dose whole-body HZE (Fe) and gamma irradiation." Radiat Res. 2016 Dec;186(6):614-23. http://dx.doi.org/10.1667/RR14530.1 ; PubMed PMID: 27905869; PubMed Central PMCID: PMC5240657 , Dec-2016
Articles in Peer-reviewed Journals	Hinkle JJ, Olschowka JA, Love TM, Williams JP, O'Banion MK. "Cranial irradiation mediated spine loss is sex-specific and complement receptor-3 dependent in male mice." Sci Rep. 2019 Dec 11;9(1):18899. https://doi.org/10.1038/s41598-019-55366-6 ; PubMed PMID: 31827187; PubMed Central PMCID: PMC6906384 , Dec-2019
Articles in Peer-reviewed Journals	Owlett L, Belcher EK, Dionisio-Santos DA, Williams JP, Olschowka JA, O'Banion MK. "Space radiation does not alter amyloid or tau pathology in the 3xTg mouse model of Alzheimer's disease." Life Sci Space Res. 2020 Nov;27:89-98. Epub2020 Aug 3. https://doi.org/10.1016/j.lssr.2020.08.001 , Nov-2020